Without ThreadPool

```
class FoodOrder implements Runnable {

    private int orderNumber;

    FoodOrder(int orderNumber) {
        this.orderNumber = orderNumber;
    }

    @Override
    public void run() {

        System.out.println(
            Thread.currentThread().getName()
            + " is processing Order " + orderNumber
        );

        //menu preparation time
        try {
            Thread.sleep(2000);
        } catch (InterruptedException e) {
            e.printStackTrace();
        }

        System.out.println(
            Thread.currentThread().getName()
            + " completed Order " + orderNumber
        );
    }
}

public class WithoutThreadPool {

    public static void main(String[] args) {

        // Create 6 Order objects - 6 tasks
      FoodOrder order1 = new FoodOrder(1);
      FoodOrder order2 = new FoodOrder(2);
      FoodOrder order3 = new FoodOrder(3);
      FoodOrder order4 = new FoodOrder(4);
      FoodOrder order5 = new FoodOrder(5);
      FoodOrder order6 = new FoodOrder(6);

        // Create 6 Thread objects
        Thread t1 = new Thread(order1);
        Thread t2 = new Thread(order2);
        Thread t3 = new Thread(order3);
        Thread t4 = new Thread(order4);
```

```
        Thread t5 = new Thread(order5);
        Thread t6 = new Thread(order6);

        // Start the threads
        t1.start();
        t2.start();
        t3.start();
        t4.start();
        t5.start();
        t6.start();
    }
}
```

With Thread Pool

```
import java.util.concurrent.ExecutorService;
import java.util.concurrent.Executors;
class Order implements Runnable {
   private int orderNumber;
   Order(int orderNumber) {
      this.orderNumber = orderNumber;
   }
   @Override
   public void run() {
      System.out.println(
         Thread.currentThread().getName()
         + " is processing Order " + orderNumber
      );

      try {
         Thread.sleep(2000);
      } catch (InterruptedException e) {
         e.printStackTrace();
      }
```

```
        System.out.println(
            Thread.currentThread().getName()
            + " completed Order " + orderNumber
        );
    }
}

public class FoodDeliveryMain {

    public static void main(String[] args) {
                                        //3 threads
        ExecutorService pool = Executors.newFixedThreadPool(3);

        Runnable runnable  = new Order(3);
        // Create and submit 6 orders
        pool.submit(new Order(1));//task1
        pool.submit(new Order(2));//task2
        pool.submit(new Order(3));//task3
        pool.submit(new Order(4));//task4
        pool.submit(new Order(5));//task5
        pool.submit(new Order(6));//task6

        // Stop accepting new tasks
        pool.shutdown();
    }
}
```